

Lab 2 GE05015: Scalar solver implementation

Modelling wind and dispersion in urban environments

Learning Objectives:

After this class, you will be able to:

- How to implement a scalar solver for simpleFoam
- How to run it

Activities:

This class is hands-on, so you will need to follow my steps with your own computer, so no activities will be carried out, but I will still ask you questions....

What we will learn might be very new to some, and not so new to others, if you have any doubt it is very important to solve it asap, so you don't get lost

Activities:

This class is hands-on, so you will need to follow my steps with your own computer, so no activities will be carried out, but I will still ask you questions....

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Remember you will use what I show you today, to run your own case for the assignment!!!

Outline of the class:

- 1.** Where are the solvers?
- 2.** Files within the solver
- 3.** Compiling
- 4.** Modifying the solver
- 5.** Setting-up a case with dispersion

Where are the solvers?

- All solvers are located within \$FOAM_SOLVERS path

```
openfoam2412:~/  
claragarciasan$ cd $FOAM_SOLVERS/  
openfoam2412:/Volumes/OpenFOAM-v2412/applications/solvers/  
claragarciasan$
```

- In our case we focus only in incompressible solvers:

```
openfoam2412:/Volumes/OpenFOAM-v2412/applications/solvers/incompressible/  
claragarciasan$ ls  
adjointOptimisationFoam      boundaryFoam      incompressibleSolversDoc.H  pimpleFoam      shallowWaterFoam  
adjointShapeOptimizationFoam  icoFoam          nonNewtonianIcoFoam        pisoFoam        simpleFoam  
openfoam2412:/Volumes/OpenFOAM-v2412/applications/solvers/incompressible/  
claragarciasan$
```

```
claragarciasan$ cd simpleFoam/  
openfoam2412:/Volumes/OpenFOAM-v2412/applications/solvers/incompressible/simpleFoam/  
claragarciasan$ ls  
createFields.H      Make      overSimpleFoam      pEqn.H      porousSimpleFoam      simpleFoam.C      SRFSimpleFoam      UEqn.H  
openfoam2412:/Volumes/OpenFOAM-v2412/applications/solvers/incompressible/simpleFoam/  
claragarciasan$
```

Files within the solver

- Inside the simpleFoam solver we have different files and folders:

h createFields.H	20 Dec 2024 at 18:00	798 bytes	C Head...Source
> Make	20 Dec 2024 at 18:00	--	Folder
> overSimpleFoam	20 Dec 2024 at 18:00	--	Folder
h pEqn.H	20 Dec 2024 at 18:00	1 KB	C Head...Source
> porousSimpleFoam	20 Dec 2024 at 18:00	--	Folder
c simpleFoam.C	20 Dec 2024 at 18:00	4 KB	C Source
> SRFSimpleFoam	20 Dec 2024 at 18:00	--	Folder
h UEqn.H	20 Dec 2024 at 18:00	423 bytes	C Head...Source

Files within the solver

- Inside the simpleFoam solver we have different files and folders:

h createFields.H	20 Dec 2024 at 18:00	798 bytes	C Head...Source
> Make	20 Dec 2024 at 18:00	--	Folder
> overSimpleFoam	20 Dec 2024 at 18:00	--	Folder
h pEqn.H	20 Dec 2024 at 18:00	1 KB	C Head...Source
> porousSimpleFoam	20 Dec 2024 at 18:00	--	Folder
c simpleFoam.C	20 Dec 2024 at 18:00	4 KB	C Source
> SRFSimpleFoam	20 Dec 2024 at 18:00	--	Folder
h UEqn.H	20 Dec 2024 at 18:00	423 bytes	C Head...Source

- We first copy the solver in our own directory, ensuring the same path as that of OF installation:

```
openfoam2412:~/OpenFOAM/claragarciasan-v2412/run/
claragarciasan$ cd $FOAM_RUN/
openfoam2412:~/OpenFOAM/claragarciasan-v2412/run/
claragarciasan$ ls
windAroundBuildings
openfoam2412:~/OpenFOAM/claragarciasan-v2412/run/
claragarciasan$ cd ..
openfoam2412:~/OpenFOAM/claragarciasan-v2412/
claragarciasan$ ls
run
openfoam2412:~/OpenFOAM/claragarciasan-v2412/
claragarciasan$ mkdir -p applications/solvers/incompressible
openfoam2412:~/OpenFOAM/claragarciasan-v2412/
claragarciasan$ cd applications/solvers/incompressible/
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/
claragarciasan$ cp -r $FOAM_SOLVERS/incompressible/simpleFoam .
```

Files within the solver

- Inside the simpleFoam solver we have different files and folders:

h	createFields.H	20 Dec 2024 at 18:00	798 bytes	C Head...Source
>	Make	20 Dec 2024 at 18:00	--	Folder
>	overSimpleFoam	20 Dec 2024 at 18:00	--	Folder
h	pEqn.H	20 Dec 2024 at 18:00	1 KB	C Head...Source
>	porousSimpleFoam	20 Dec 2024 at 18:00	--	Folder
c	simpleFoam.C	20 Dec 2024 at 18:00	4 KB	C Source
>	SRFSimpleFoam	20 Dec 2024 at 18:00	--	Folder
h	UEqn.H	20 Dec 2024 at 18:00	423 bytes	C Head...Source

- Now we can delete those files that we do not need:

```
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/simpleFoam/
claragarciasan$ rm -r porousSimpleFoam/ SRFSimpleFoam/ overSimpleFoam/
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/simpleFoam/
claragarciasan$
```

- Now we can rename all folders and files that are necessary within the solver to the new solver name:

```
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/simpleFoam/
claragarciasan$ ls
createFields.H  Make                pEqn.H             simpleFoam.C       UEqn.H
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/simpleFoam/
claragarciasan$ mv simpleFoam.C scalarSimpleFoam.C
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/simpleFoam/
claragarciasan$ cd ..
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/
claragarciasan$ mv simpleFoam/ scalarSimpleFoam
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/
claragarciasan$
```

Files within the solver

- Inside the simpleFoam solver we have different files and folders:

h createFields.H	20 Dec 2024 at 18:00	798 bytes	C Head...Source
> Make	20 Dec 2024 at 18:00	--	Folder
> overSimpleFoam	20 Dec 2024 at 18:00	--	Folder
h pEqn.H	20 Dec 2024 at 18:00	1 KB	C Head...Source
> porousSimpleFoam	20 Dec 2024 at 18:00	--	Folder
c simpleFoam.C	20 Dec 2024 at 18:00	4 KB	C Source
> SRFSimpleFoam	20 Dec 2024 at 18:00	--	Folder
h UEqn.H	20 Dec 2024 at 18:00	423 bytes	C Head...Source

- Files after arranging and cleaning:

▼ scalarSimpleFoam	Today at 08:10	--	Folder
h createFields.H	Today at 08:06	798 bytes	C Head...Source
> Make	Today at 08:06	--	Folder
h pEqn.H	Today at 08:06	1 KB	C Head...Source
c scalarSimpleFoam.C	Today at 08:06	4 KB	C Source
h UEqn.H	Today at 08:06	423 bytes	C Head...Source

- What are we still missing???

Files within the solver

- What are we still missing??? THE CEqn.H!!!

```
claragarciasan$ vi $FOAM_SOLVERS/basic/scalarTransportFoam/scalarTransportFoam.C
```

```
while (simple.correctNonOrthogonal())
{
    fvScalarMatrix TEqn
    (
        fvm::ddt(T)
        + fvm::div(phi, T)
        - fvm::laplacian(DT, T)
        ==
        fvOptions(T)
    );

    TEqn.relax();
    fvOptions.constrain(TEqn);
    TEqn.solve();
    fvOptions.correct(T);
}
```

```
claragarciasan$ vi CEqn.H
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/
```

- We copy the previous lines there!

Compiling

- Before we move forward with implementation, we will start compiling our solver, just with the change in name

h CEqn.H	Today at 08:16	389 bytes	C Head...Source
h createFields.H	Today at 08:06	798 bytes	C Head...Source
▼ Make	Today at 08:06	--	Folder
files	Today at 08:06	46 bytes	Document
options	Today at 08:06	704 bytes	Document
h pEqn.H	Today at 08:06	1 KB	C Head...Source
c scalarSimpleFoam.C	Today at 08:06	4 KB	C Source
h UEqn.H	Today at 08:06	423 bytes	C Head...Source

simpleFoam.C

```
EXE = $(FOAM_APPBIN)/simpleFoam
```



scalarSimpleFoam.C

```
EXE = $(FOAM_USER_APPBIN)/scalarSimpleFoam
```

- Then we can compile: wmake

```
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/scalarSimpleFoam/
claragarciasan$ wmake
Making dependencies: scalarSimpleFoam.C
clang++ -std=c++17 -m64 -pthread -ftrapping-math -ffp-contract=off -DOPENFOAM=2412 -DWM_DP -DWM_LABEL_SIZE=32 -Wall -Wextra -Wold-style-cast -Wnon-virtual-dtor -Wno-unused-parameter -Wno-invalid-offsetof -Wno-undef
ined-var-template -Wno-unknown-warning-option -O3 -DNoRepository -ftemplate-depth-100 -I/Volumes/OpenFOAM-v2412/env/include -I/Volumes/OpenFOAM-v2412/src/finiteVolume/lnInclude -I/Volumes/OpenFOAM-v2412/src/meshT
ools/lnInclude -I/Volumes/OpenFOAM-v2412/src/sampling/lnInclude -I/Volumes/OpenFOAM-v2412/src/TurbulenceModels/turbulenceModels/lnInclude -I/Volumes/OpenFOAM-v2412/src/TurbulenceModels/incompressible/lnInclude -I/V
olumes/OpenFOAM-v2412/src/transportModels -I/Volumes/OpenFOAM-v2412/src/transportModels/incompressible/singlePhaseTransportModel -I/Volumes/OpenFOAM-v2412/src/dynamicMesh/lnInclude -I/Volumes/OpenFOAM-v2412/src/dyn
amicFvMesh/lnInclude -iquote. -IlnInclude -I/Volumes/OpenFOAM-v2412/src/OpenFOAM/lnInclude -I/Volumes/OpenFOAM-v2412/src/OSspecific/POSIX/lnInclude -fPIC -c scalarSimpleFoam.C -o Make/darwin64ClangDPInt320pt/scal
arSimpleFoam.o
clang++ -std=c++17 -m64 -pthread -ftrapping-math -ffp-contract=off -DOPENFOAM=2412 -DWM_DP -DWM_LABEL_SIZE=32 -Wall -Wextra -Wold-style-cast -Wnon-virtual-dtor -Wno-unused-parameter -Wno-invalid-offsetof -Wno-undef
ined-var-template -Wno-unknown-warning-option -O3 -DNoRepository -ftemplate-depth-100 -I/Volumes/OpenFOAM-v2412/env/include -I/Volumes/OpenFOAM-v2412/src/finiteVolume/lnInclude -I/Volumes/OpenFOAM-v2412/src/meshT
ools/lnInclude -I/Volumes/OpenFOAM-v2412/src/sampling/lnInclude -I/Volumes/OpenFOAM-v2412/src/TurbulenceModels/turbulenceModels/lnInclude -I/Volumes/OpenFOAM-v2412/src/TurbulenceModels/incompressible/lnInclude -I/V
olumes/OpenFOAM-v2412/src/transportModels -I/Volumes/OpenFOAM-v2412/src/transportModels/incompressible/singlePhaseTransportModel -I/Volumes/OpenFOAM-v2412/src/dynamicMesh/lnInclude -I/Volumes/OpenFOAM-v2412/src/dyn
amicFvMesh/lnInclude -iquote. -IlnInclude -I/Volumes/OpenFOAM-v2412/src/OpenFOAM/lnInclude -I/Volumes/OpenFOAM-v2412/src/OSspecific/POSIX/lnInclude -fPIC -rpath @executable_path/../lib/openmpi -rpath @executable
_path/../lib -rpath @executable_path/../lib/dummy -rpath @executable_path/../lib/../../env/lib -WL,-execute,-undefined,dynamic_lookup -L/Volumes/OpenFOAM-v2412/env/lib Make/darwin64ClangDPInt320pt/scalarSimpleF
oam.o -L/Volumes/OpenFOAM-v2412/platforms/darwin64ClangDPInt320pt/lib \
-lfiniteVolume -lfvOptions -lmeshTools -lsampling -ldynamicMesh -ldynamicFvMesh -lturbulenceModels -lincompressibleTurbulenceModels -lincompressibleTransportModels -latmosphericModels -lOpenFOAM -ldl \
-lm -o /Users/claragarciasan/OpenFOAM/claragarciasan-v2412/platforms/darwin64ClangDPInt320pt/bin/scalarSimpleFoam
openfoam2412:~/OpenFOAM/claragarciasan-v2412/applications/solvers/incompressible/scalarSimpleFoam/
claragarciasan$
```

Modifying the solver

- Now let's add all the little bits we need!
- CEqn.H in scalarSimpleFoam.C and C variable in createFields.H

```
Info<< "\nStarting time loop\n" << endl;

while (simple.loop())
{
    Info<< "Time = " << runTime.timeName() << nl << endl;

    // Do any mesh changes
    mesh.controlledUpdate();

    if (mesh.changing())
    {
        MRF.update();
    }

    // --- Pressure-velocity SIMPLE corrector
    {
        #include "UEqn.H"
        #include "pEqn.H"
        #include "CEqn.H"
    }

    laminarTransport.correct();
    turbulence->correct();

    runTime.write();

    runTime.printExecutionTime(Info);
}
```

```
Info<< "Reading field p\n" << endl;
volScalarField p
(
    IOobject
    (
        "p",
        runTime.timeName(),
        mesh,
        IOobject::MUST_READ,
        IOobject::AUTO_WRITE
    ),
    mesh
);

Info<< "Reading field C\n" << endl;
volScalarField C
(
    IOobject
    (
        "C",
        runTime.timeName(),
        mesh,
        IOobject::MUST_READ,
        IOobject::AUTO_WRITE
    ),
    mesh
);
```

Modifying the solver

- Now let's add all the little bits we need!

```
singlePhaseTransportModel laminarTransport(U, phi);

autoPtr<incompressible::turbulenceModel> turbulence
(
    incompressible::turbulenceModel::New(U, phi, laminarTransport)
);

Info<< "Reading transportProperties\n" << endl;

IOdictionary transportProperties
(
    IOobject
    (
        "transportProperties",
        runTime.constant(),
        mesh,
        IOobject::MUST_READ_IF_MODIFIED,
        IOobject::NO_WRITE
    )
);

Info<< "Reading diffusivity DT\n" << endl;

dimensionedScalar DT("DT", dimViscosity, transportProperties);
```

-
- Add transportProperties to input diffusion term in createFields.H

Modifying the solver

- Change DC by Schmidt numbers:

```
Info<< "Reading diffusivity DT\n" << endl;  
dimensionedScalar DT("DT", dimViscosity, transportProperties);
```



```
Info<< "Reading Schmidt number Sc\n" << endl;  
dimensionedScalar Sc("Sc", dimless, transportProperties);  
  
Info<< "Reading Turbulent Schmidt number Sct\n" << endl;  
dimensionedScalar Sct("Sct", dimless, transportProperties);
```

- Modify CEqn.H
- Change all naming to C, since our variable is C!

```
volScalarField DC("DC", turbulence-> nu()/Sc+turbulence-> nut()/Sct);  
  
while (simple.correctNonOrthogonal())  
{  
    fvScalarMatrix CEqn  
    (  
        fvm::ddt(C)  
        + fvm::div(phi, C)  
        - fvm::laplacian(DC, C)  
        ==  
        fvOptions(C)  
    );  
  
    CEqn.relax();  
    fvOptions.constrain(CEqn);  
    CEqn.solve();  
    fvOptions.correct(C);  
}
```

Setting up a case with dispersion

- Modify system/fvSchemes, system/fvSolution

```
gradSchemes
{
    default          Gauss linear;

    limited          cellLimited Gauss linear 1;
    grad(U)          $limited;
    grad(k)          $limited;
    grad(epsilon)    $limited;
    grad(C)          $limited;
}

divSchemes
{
    default          none;

    div(phi,U)       bounded Gauss linearUpwind limited;

    turbulence        bounded Gauss limitedLinear 1;
    div(phi,k)        $turbulence;
    div(phi,epsilon)  $turbulence;
    div(phi,C)        $turbulence;

    div((nuEff*dev2(T(grad(U)))) Gauss linear;
}
```

```
solvers
{
    p
    {
        solver          GAMG;
        smoother        GaussSeidel;
        tolerance       1e-6;
        relTol          0.1;
    }

    "(U|k|omega|epsilon|C)"
    {
        solver          smoothSolver;
        smoother        symGaussSeidel;
        tolerance       1e-6;
        relTol          0.1;
    }
}

SIMPLE
{
    residualControl
    {
        p              1e-4;
        U              1e-4;
        "(k|omega|epsilon|C)" 1e-4;
    }
    nNonOrthogonalCorrectors 0;
    pRefCell            0;
    pRefValue            0;
}

potentialFlow
{
    nNonOrthogonalCorrectors 10;
}

relaxationFactors
{
    fields
    {
        p              0.3;
    }
    equations
    {
        U              0.7;
        "(k|omega|epsilon|C).*" 0.7;
    }
}
```

Setting up a case with dispersion

- Add O/C field

```
/*----- C++ -----*\
=====
\\ / F i e l d      | OpenFOAM: The Open Source CFD Toolbox
\\ / O peration    | Website: https://openfoam.org
\\ / A nd          | Version: 7
\\ \ M anipulation |
\*-----*/

FoamFile
{
    version      2.0;
    format       ascii;
    class        volScalarField;
    object       C;
}
// *****

cInlet          0;    // approx k = 1.5*(I*U)^2 ; I = 0.1

dimensions      [0 0 0 0 0 0];

internalField   uniform $cInlet;

boundaryField
{
    inlet
    {
        type      fixedValue;
        value      uniform $cInlet;
    }

    outlet
    {
        type      inletOutlet;
        inletValue uniform $cInlet;
        value      uniform $cInlet;
    }

    wall
    {
        type      zeroGradient;
    }

    #includeEtc "caseDicts/setConstraintTypes"
}

// *****
```

Setting up a case with dispersion

- Add constant/fvOptions
- This file injects the concentration of the pollutant in a determined location

```
/*-----* C++ *-----*\
=====
\\      / F i e l d      | OpenFOAM: The Open Source CFD Toolbox
\\      / O p e r a t i o n | Website: https://openfoam.org
\\      / A n d             | Version: 7
  \\    / M a n i p u l a t i o n |
\*-----*/

FoamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    location     "constant";
    object       fvOptions;
}

// *****

injector1
{
    timeStart     0;
    duration      3000;
    selectionMode  points;
    points
    (
        (55 62 2)
    );
}

options
{
    CSource1
    {
        type          scalarSemiImplicitSource;

        $injector1;

        volumeMode     specific;
        injectionRateSuSp
        {
            C          (3000 0); // kg/m3
        }
    }
}

// *****
```